Lecture 7(c): Electric Vehicle FQ

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Joint work with Leander Heulens, Alexandre Gouveia

Agenda

- Electric vehicle growth in distribution networks
- Flexibility quantification method for EV
- Mathematical model for FQ
- Numerical case studies
- Conclusions

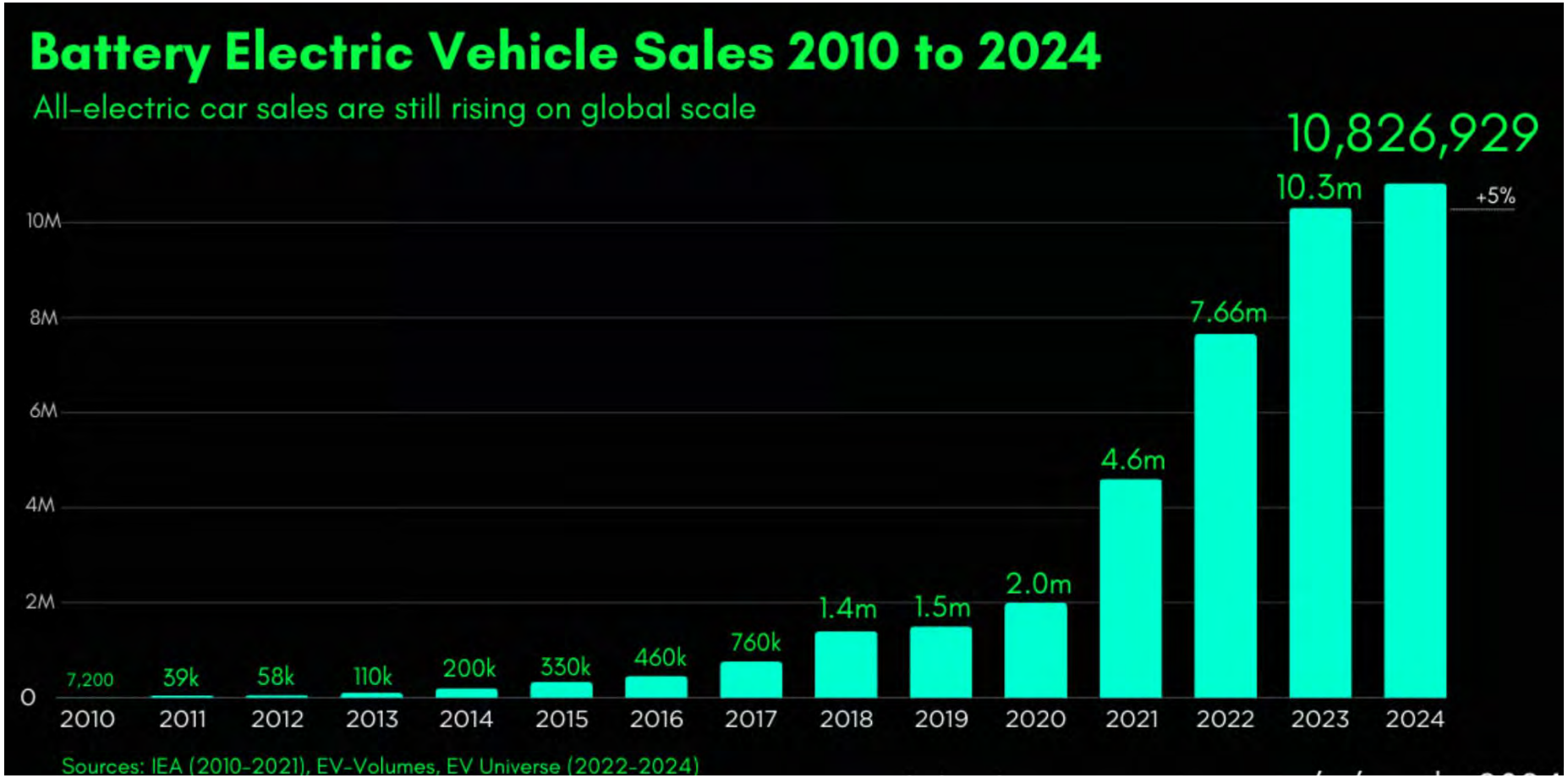
Electric vehicle growth in distribution networks

Context

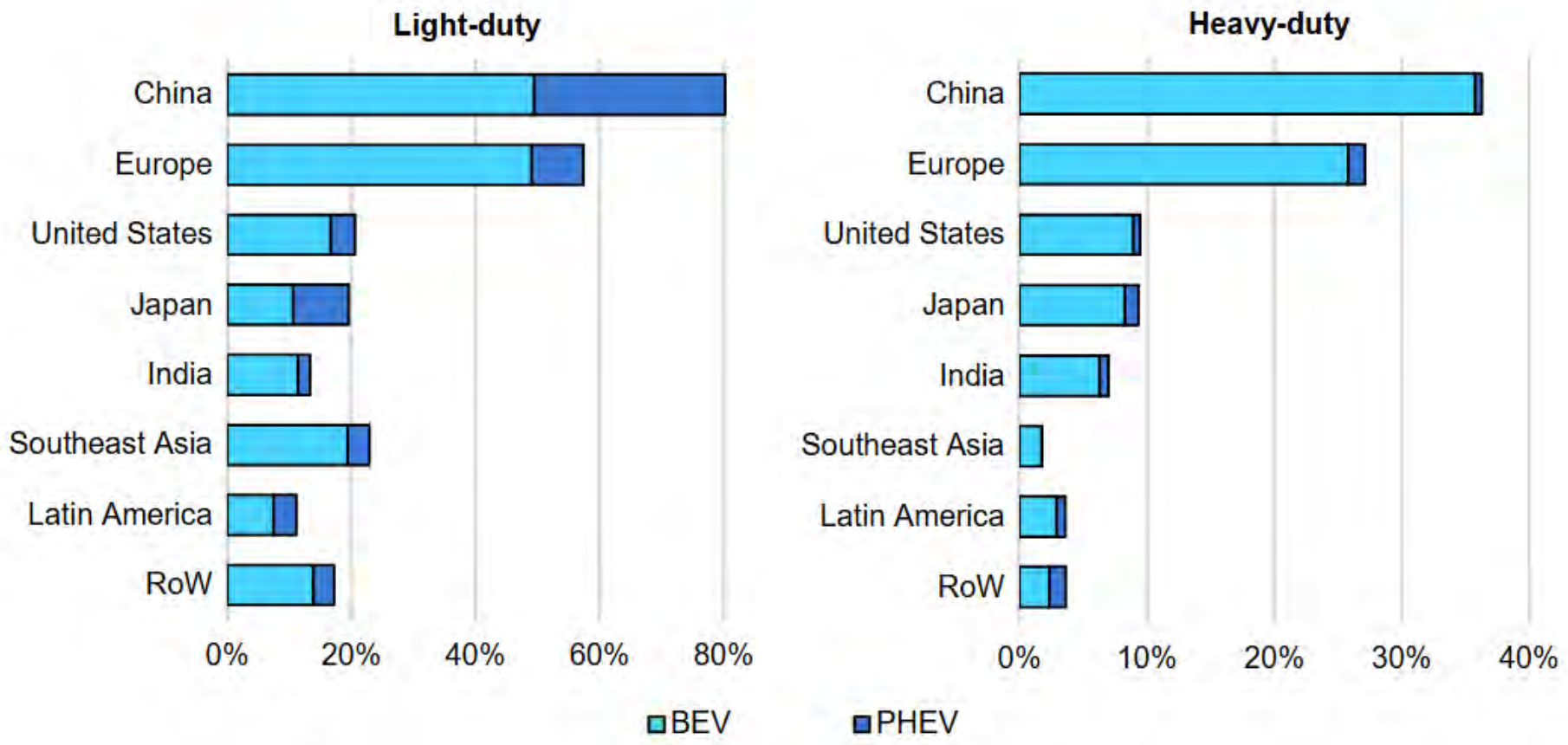
- EU's Sustainable and Smart Mobility Strategy targets 30 million zero-emission cars by 2030
- Across the EU and the UK, approximately 2 million battery-electric cars were registered in 2024, accounting for 15.4 % of new sales
- EU aims to ban all carbon-emitting vehicles by 2035

Sources:
[Ref 1]. https://transport.ec.europa.eu/transport-themes/mobility-strategy_en
[Ref 2]. https://evwire.com/p/evsales2024
[Ref 3]. https://ec.europa.eu/commission/presscorner/detail/en/ip_22_6462

EV growth trend



Electric vehicle sales share by mode and region in the Stated Policies Scenario, 2030



BEV = battery electric vehicle; PHEV = plug-in hybrid electric vehicle; RoW = Rest of World. Light-duty refers to passenger light-duty vehicles and light commercial vehicles. Heavy-duty refers to buses, medium-duty and heavy-duty trucks

Source: Global EV outlook, https://iea.blob.core.windows.net/assets/7ea38b60-3033-42a6-9589-71134f4229f4/GlobalEVOutlook2025.pdf

Electrification of transportation

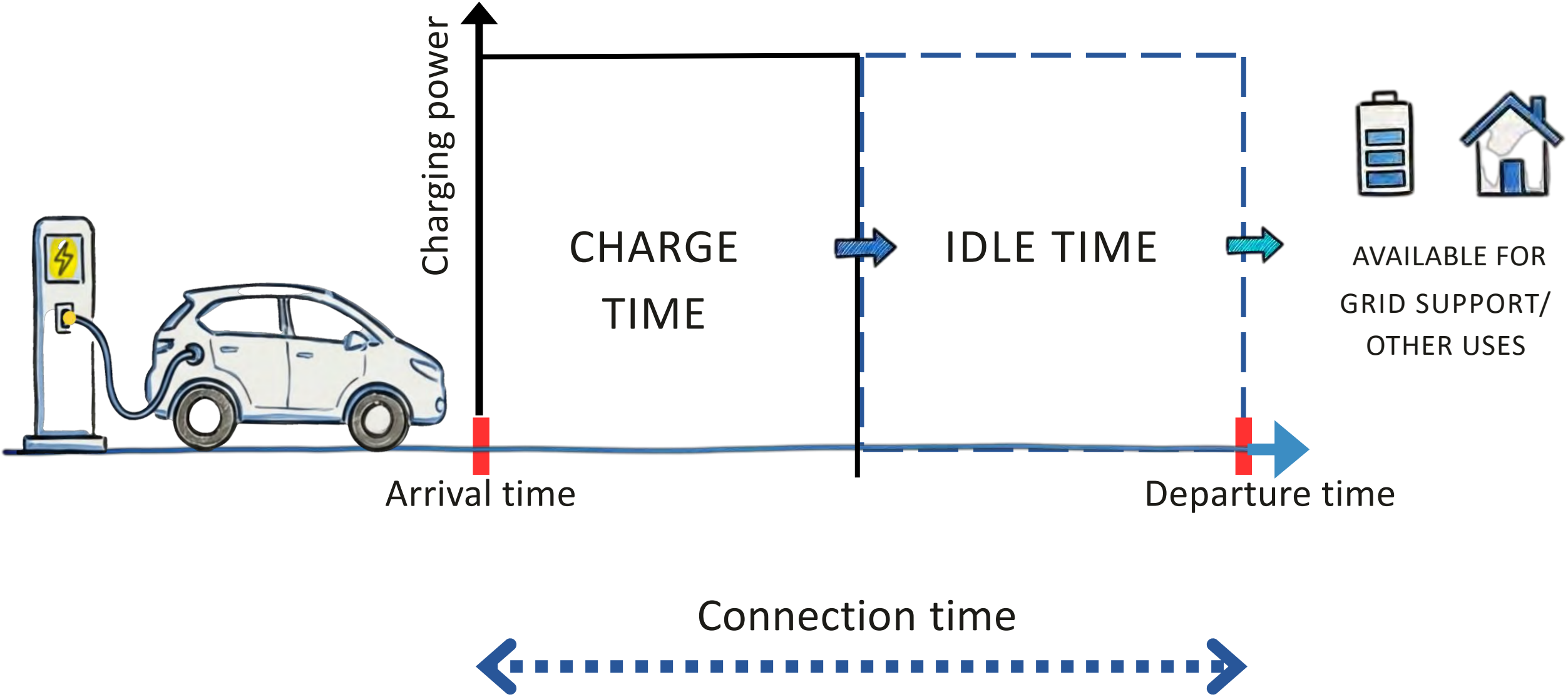
- Transportation sector is largely getting electrified → adding a lot of new load onto the power networks
  - EV adoption is increasing the household electricity consumption by more than 20% and up to 50%.
- EV load is highly simultaneous
  - Simultaneity factor ranging from 0.1 to 0.5 (10%–50%) for residential or public charging

Power networks are getting more and more stressed

Silber, F., Scheubner, S., & Märtz, A. (2024). Analysis of the simultaneity factor of fast-charging sites using Monte-Carlo simulation. *International Journal of Electrical Power & Energy Systems*, *155*, 109540.

Flexibility quantification method for EV

NOTION OF EV FLEXIBILITY



Notion of EV flexibility

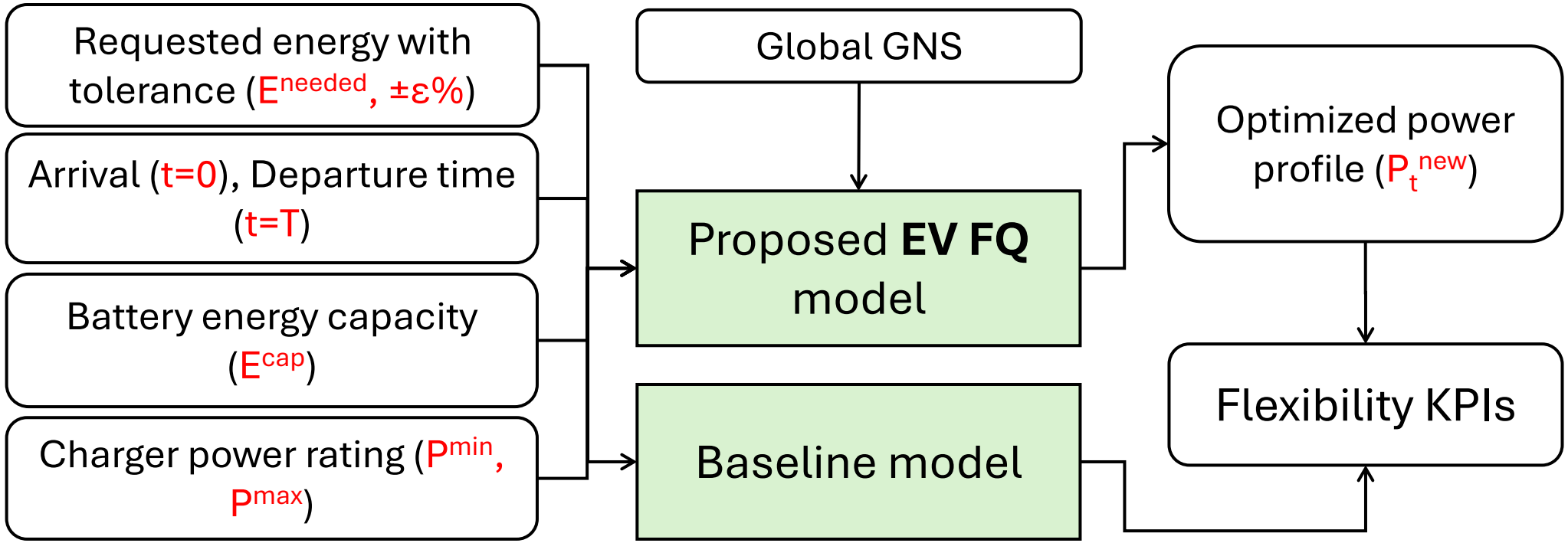
Notion of flexibility for EVs: The notion of flexibility for EVs is based on the connection time being much higher than the charge time. Connection time refers to the time between the arrival and the departure of the EV at the charging point. The charge time is the minimum time needed to charge the EV to the desired energy level (requested by the EV owner). There is no potential for flexibility if the connection time is equal to the charge time. EV flexibility relies on the electrical characteristics of the EV battery, the operating limits of the EV and information about the connection (arrival and departure) with the charger.

Mathematical model for FQ

Notation

- E^{min}, E^{max}: the minimum and the maximum battery energy (kWh) at departure time T .
- E^{init}: the battery energy (kWh) at the start of the charging interval at t = 0.
- E^{cap}: the energy capacity of the battery (kWh).
- η_{ch}, η_{dis}: charging and discharging efficiencies
- E^{needed}: denotes the requested kWh for charging. The requested energy is frequently also denoted as state-of-charge, given as $\rho = E^{needed}/E^{cap}$.

FQ framework



Battery process constraint

$$E_t = E_{t-1} + \Delta t \left(\eta_{\text{ch}} \cdot \max\{P_t^{\text{new}}, 0\} - \frac{\max\{-P_t^{\text{new}}, 0\}}{\eta_{\text{dis}}} \right), \quad \forall t \in T^{\text{ch}}$$

$$\delta_t^{ch} = \max\{P_t^{\text{new}}, 0\} \qquad \delta_t^{dis} = \max\{-P_t^{\text{new}}, 0\}$$

$$E_{t+1} = E_t + \Delta t \left(\eta_{\text{ch}} \cdot \delta_t^{ch} - \frac{\delta_t^{dis}}{\eta_{\text{dis}}} \right), \quad \forall t \in T^{\text{ch}}$$

$$0 \leq E_t \leq E^{\text{cap}}, \quad \forall t \in T^{\text{ch}}$$

$$P_t^{\text{new}} = \delta_t^{ch} - \delta_t^{dis}, \quad \forall t \in T^{\text{ch}}$$

$$\delta_t^{ch} \leq M_2 \, z_t, \quad \forall t \in T^{\text{ch}}$$

$$P^{\min} \leq P_t^{\text{new}} \leq P^{\max},$$

$$\delta_t^{dis} \leq M_2 \, (1 - z_t), \quad \forall t \in T^{\text{ch}}$$

Comfort constraint → Deadline constraint

- Battery capacity level at the end of charging session

$$E_T \in [E^{\min}, E^{\max}]$$

$$E^{\min} = (1-\epsilon)(E^{\text{init}} + E^{\text{needed}})$$

$$E^{\max} = (1+\epsilon)(E^{\text{init}} + E^{\text{needed}})$$

Objective function for EV FQ

- Maximize incentive of providing flexibility
- Minimize cycles of operation of the battery

$$\max_{P_t^{\text{new}}} \Delta t \sum_{t \in T^{\text{ch}}} \left(\lambda_t^{\text{up}} \delta_t^{+} + \lambda_t^{\text{down}} \delta_t^{-} - \lambda^{\text{penalty}} \left(\delta_t^{\text{ch}} + \delta_t^{\text{dis}} \right) \right)$$

Modeling for V1G and V2G

- V1G → EV can only charge from the grid
  - Set lower limit of power: P^{min} = 0kW
- V2G → EV can both charge from and discharge into the grid

- Impact of baseline in EV FQ
  - Without baseline
    - V1G → can only provide ramp up flexibility
    - V2G → can provide both ramp up and down flexibility
  - With baseline

Numerical case study

Case study: Exemplary case

- Battery parameters
  - $P^{max} = -P^{min} = 11.1$ kW
  - $\eta_{ch} = \eta_{dis} = 0.95$
  - $E^{cap} = 60$ kWh

- Requested energy
  - $E^{needed} = 40$ kWh
  - $T_{ch} = 10$ hours

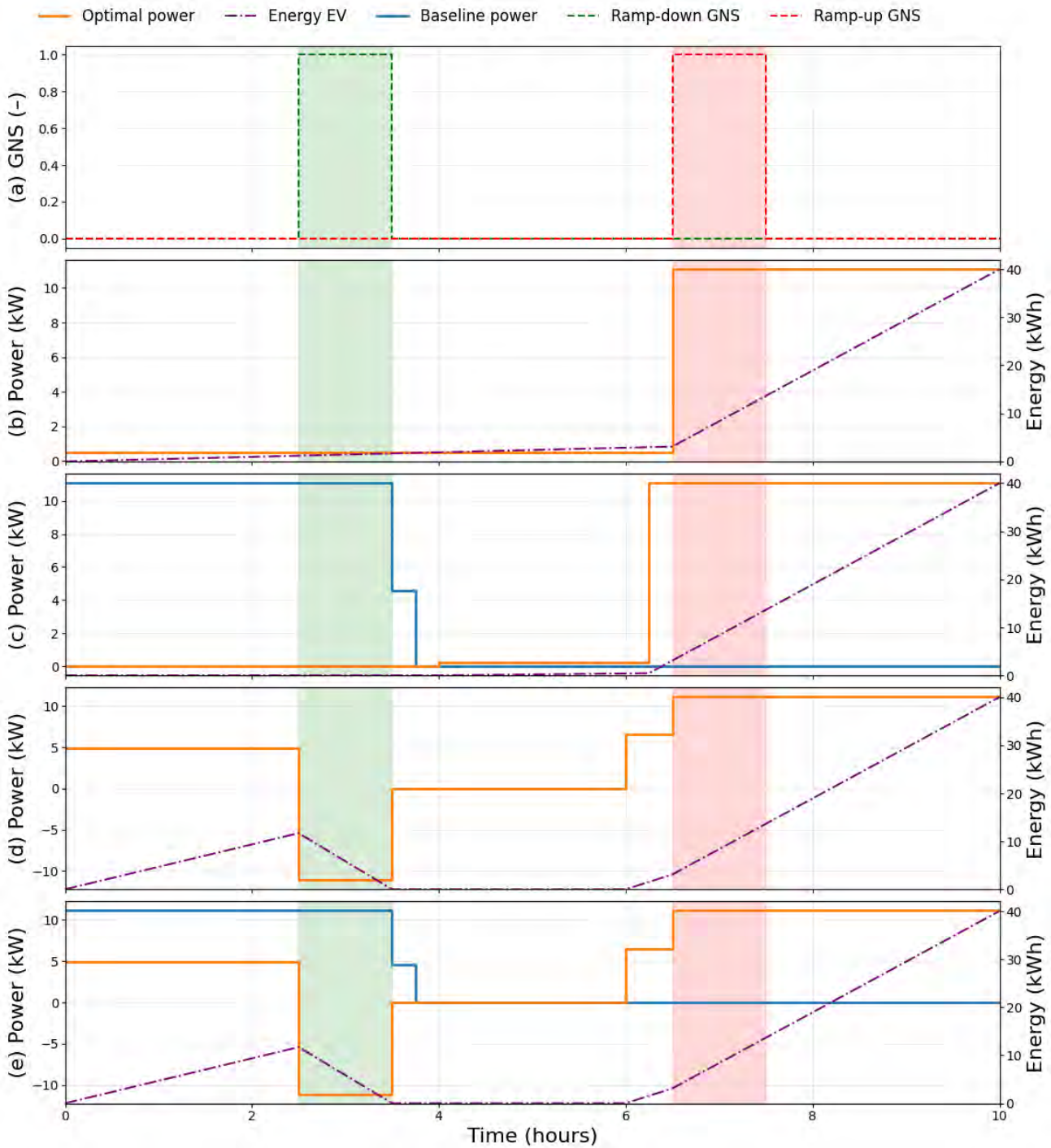
Grid needs signal

V1G without baseline

V1G with baseline

V2G without baseline

V2G with baseline



Results

Mode	Base-line	E^{flex} [kWh]	$E^{\text{flex}}_{\text{up}}$ [kWh]	$E^{\text{flex}}_{\text{down}}$ [kWh]	$\epsilon^{\text{abs}}_{\text{needed}}$	$\epsilon^{\text{flex}}_{\text{needed}}$
V1G	No	11.1	11.1	0.0	1.0	0.3
	Yes	22.2	11.1	11.1	1.0	0.6
V2G	No	22.2	11.1	11.1	1.6	0.6
	Yes	33.3	11.1	22.2	1.6	0.8

Observations:

The extra E^{flex} for V2G comes at a cost of higher value of ϵ^{abs}_{needed} which increases from 1 to 1.6 for V1G toV2G modes

Conclusions

Conclusions

- EV can provide flexibility based on the idle time
- Inclusion of baseline affects the amount of flexibility
- The quantified flexibility is much lower than requested energy for EV charging

Thank you for your attention!

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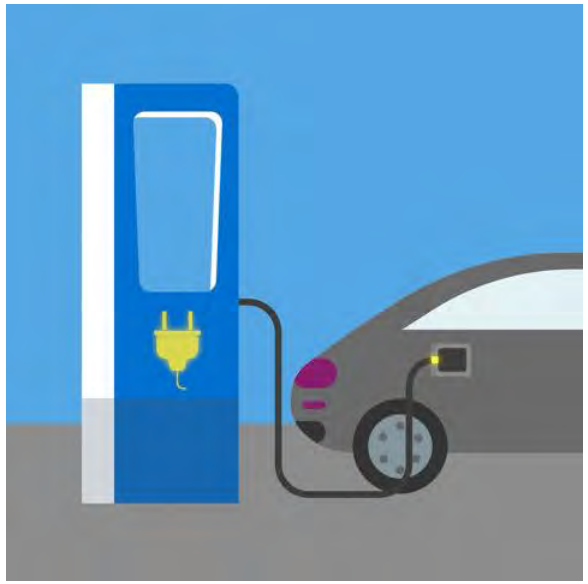
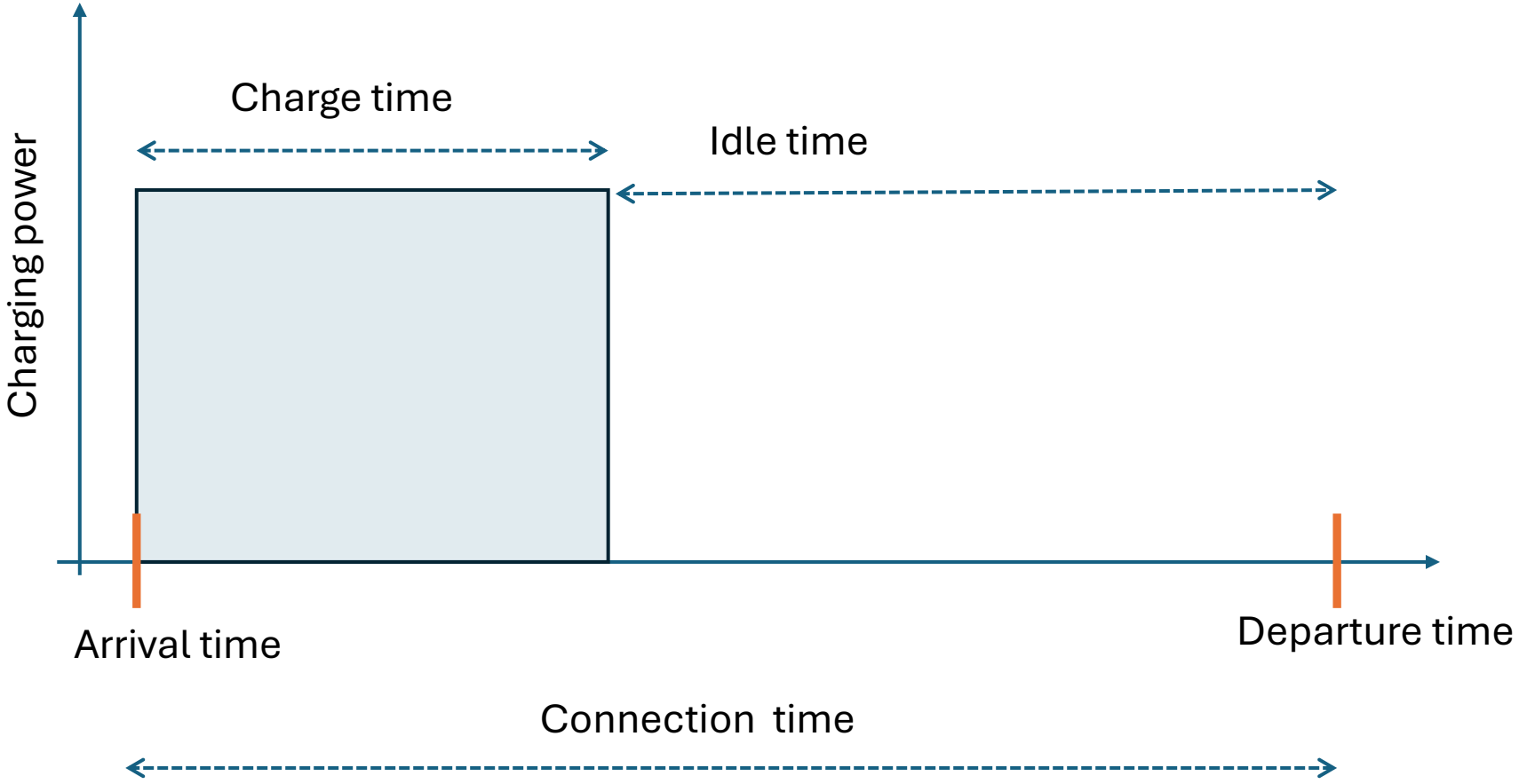
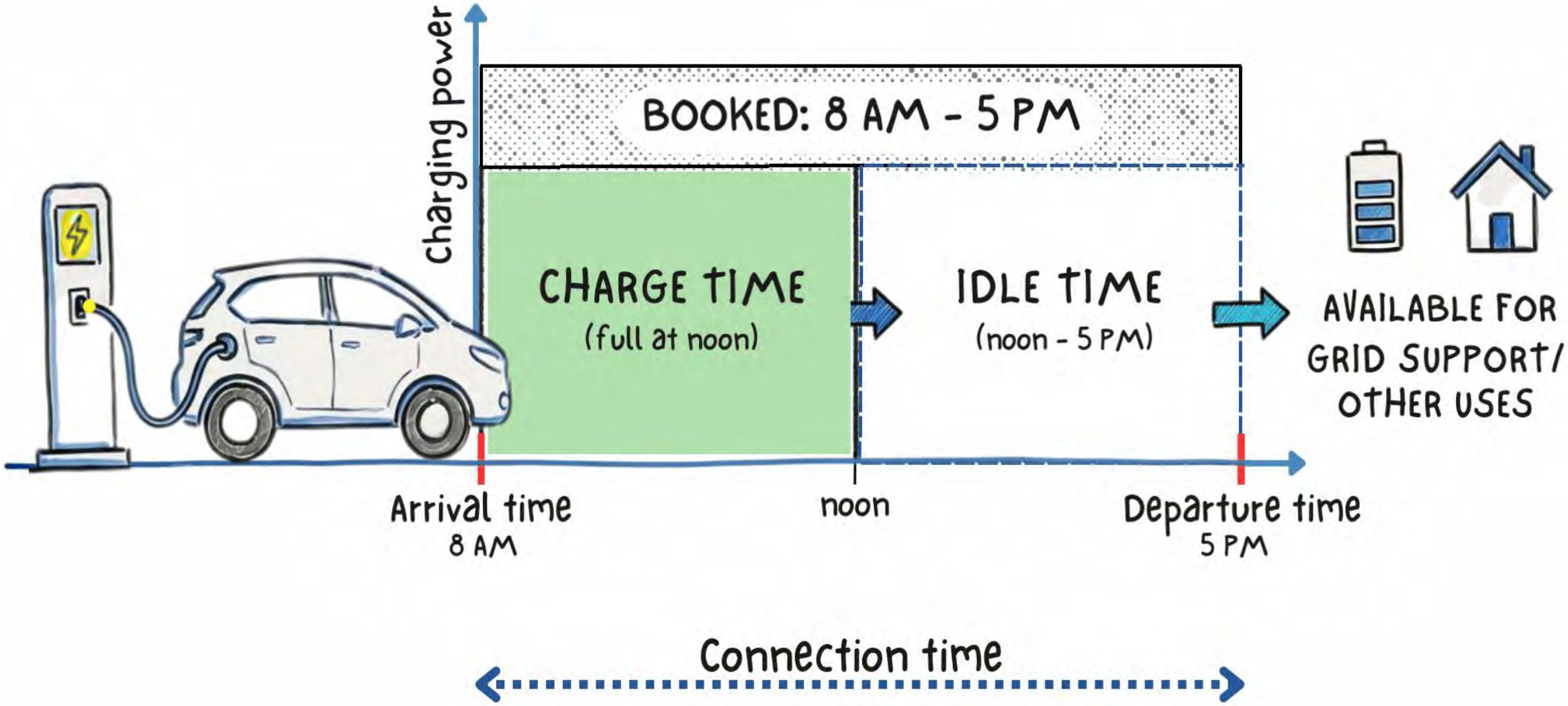
Notion of EV flexibility



Source: https://www.aral.de/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